

Week 04

01) Confidence Interval

- * Confidence Interval (CI) gives a range of values that is likely to contain the true population parameter (like the mean or proportion)
- * It is based on sample data - since you can't observe the whole population, you use your sample to estimate it.

$$CI = \text{estimate} \pm \text{margin of error}$$

- * It's about how confident you are in your estimate, not the probability that the value is in the interval

- * Common CL $\Rightarrow$ 90%, 95%, 99%

Changing the interval

- o When you change the confidence level or sample size, the interval width changes
- * Higher Confidence Level $\Rightarrow$ wider interval
- * Larger sample size $\Rightarrow$ narrower interval
- * More variability $\Rightarrow$ wider interval
(standard deviation)

2) Margin of Error

*) The MOE tells you how far your estimate might be from the true value.

$$MOE = z^* \cdot \frac{\sigma}{\sqrt{n}}$$

) z^ is the critical value which will be obtained by the confidence level

$$\frac{\text{Confidence level}}{2} = 1 - \frac{\alpha}{2}$$

α = Significance level

ex: If confidence level is 95%, then

~~1 - 0.05~~, ~~1 - 0.05~~ $\alpha = 0.05$

0.05 = 5% risk
0.1 = 10% risk
0.01 = 1% risk

~~1 - 0.05~~ $z^*_{\frac{\alpha}{2}} = 1.96$

$$z^*_{0.025} = \frac{\sigma - \mu}{\sigma} \quad \text{Cumulative prob } (\Rightarrow) 2\sigma$$

1) 95% CL $\Rightarrow \alpha = 0.05 \Rightarrow z^* = 1.96$

2) 99% CL $\Rightarrow \alpha = 0.01 \Rightarrow z^* = 2.576$

3) 90% CL $\Rightarrow \alpha = 0.10 \Rightarrow z^* = 1.645$

- * Reduce MOE by:
- i) Increasing sample size (n)
 - ii) Decreasing standard deviation (σ)
 - iii) Lowering Confidence level

3) Calculate Confidence Interval

- 1) Find the sample mean
- 2) Define a desired confidence level ($1-\alpha$)
- 3) Get the critical value ($z^*_{1-\alpha/2}$)

- 4) Find the standard error $\left(\frac{\sigma}{\sqrt{n}}\right)$

- 5) Find the MOE $\left(z^*_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right)$

- 6) Add/subtract MOE to the sample mean.

$$\text{Confidence Interval} = \bar{x} \pm z^*_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

- * Simple random sample
- * Sample size > 30 or population is approximately normal

$$\text{Sample size } (n) \geq \left(\frac{z_{1-\alpha/2} \cdot \sigma}{\text{MOE}} \right)^2$$

4) Difference between Confidence and probability

→ Confidence ⇒ how often the method captures the true value in repeated samples

"If I repeat the process 100 times, 95 of my intervals would include the true mean"

* Probability ⇒ applies before data collection, not after

Once you have your CI, the true value either is or isn't inside - there's no probability anymore.

5) Unknown Standard Deviation

- * If the population standard deviation is unknown (which is common), we use the sample standard deviation (s) and the t -distribution instead of z .

$$CI = \bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

- * As n increases, t distribution $\approx$ normal distribution

6) Confidence Intervals for Proportion

- * Used when estimating proportions (like % of people who prefer coffee)

$$CI = \hat{p} \pm z^*_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- $\hat{p}$ = sample proportion
- Works similarly to mean based CIs
- Larger $n \rightarrow$ more precise proportion estimate

Ex:

60% support rate $\rightarrow$ CI (56%, 64%) at 95% confidence.

7] Hypotheses

*] In hypothesis testing, we start by making two opposing statements about a population.

1) Null Hypothesis (H_0):

*] This is the default or "no effect" statement

Ex: "The new drug has no effect compared to the old one" (The mean = 50)

2) Alternative Hypothesis (H_1 or H_a)

*] This is what you want to prove, that there is an effect or difference

Ex: "The new drug works better than the old one" (The mean $\neq$ 50)

8) Type I Error (a error)

*] This happens when you reject the null hypothesis when it is actually true.

(You think you found an effect, but it's a false alarm)

ex:- You conclude the new drug works (reject H_0) but in reality it doesn't

Significance level

- *) This is the probability of making a Type I error is called the significance level (α)
- *) It means how much risk of a Type I Error we are willing to accept
- *) Common values: 0.05 (5%), 0.01 (1%), 0.10 (10%)
(can never be 0)

9) Type II Error (β error)

- *) This happens when you fail to reject the null hypothesis when it's actually false.
(you miss a real effect)

ex: You conclude the new drug doesn't work but in reality, it actually does (fail to reject H_0)

- *) The power of a test is $(1-\beta)$, meaning the probability of correctly rejecting a false H_0

$$\text{Standard Error} = \frac{s}{\sqrt{n}}$$

b) Right / Left / Two-tailed Tests

- * When we do hypothesis test, we check whether our sample result is extreme enough (far from expected under H_0)
- * The direction of extreme depends on the alternative hypothesis (H_1 or H_a)

Left - Tailed Test

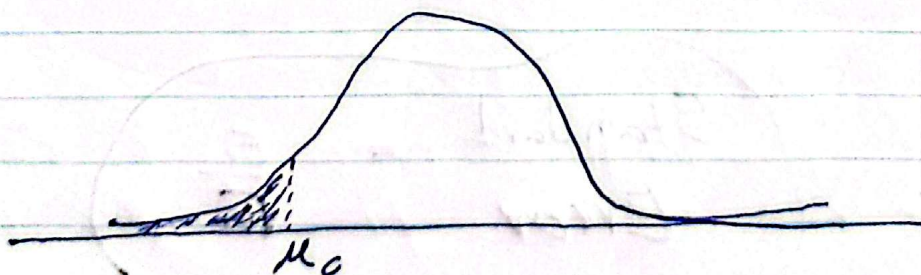
- * Alternative Hypothesis : H_1 : Parameter $<$ value
- * We test if population mean (or proportion) is less than the hypothesized value.

Ex: $H_0 : \mu = 50$

$H_1 : \mu < 50$

→ Used when we expect a decrease (The average battery life is less than advertised)

- * The rejection region is the left tail of the normal distribution



Right-Tailed Test

* Alternative Hypothesis: $H_1: \text{parameter} > \text{value}$

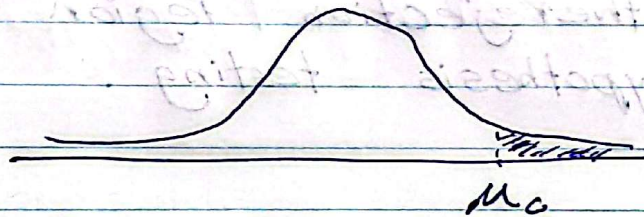
* We test if the population mean is greater than the hypothesized value

$$H_0: \mu = 50$$

$$H_1: \mu > 50$$

→ Used when we expect increase. ("The new teaching method increases test scores")

* The rejection region is the right tail of the distribution



Two-Tailed Test

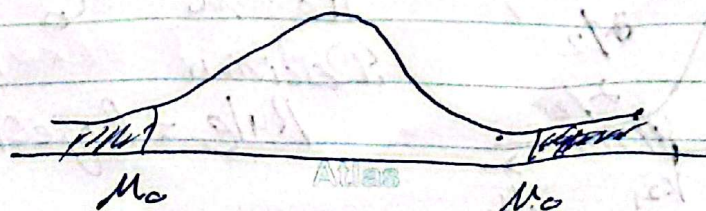
* Alternative hypothesis: $H_1: \text{parameter} \neq \text{value}$

* We test if the population mean is different than hypothesized value

$$\text{ex: } H_0: \mu = 50$$

$$H_1: \mu \neq 50$$

→ Used when we know (just want to know) if there's difference



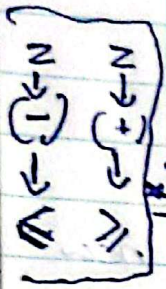
11) P-value

left	Right	Two Tail
z table according to z	$1 - (\text{z table according to } z)$	$2 \times (\text{z table according to } z)$

$p \leq \alpha$

 $\rightarrow$ Reject H_0 (Evidence supports H_0)

$p > \alpha$

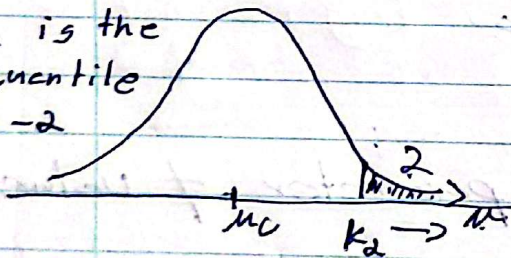
 $\rightarrow$ Fail to Reject H_0 (Not enough evidence)

p-value is the probability (under null hypo..) of getting a test statistic as extreme or more extreme than the one observed

12) Critical Values

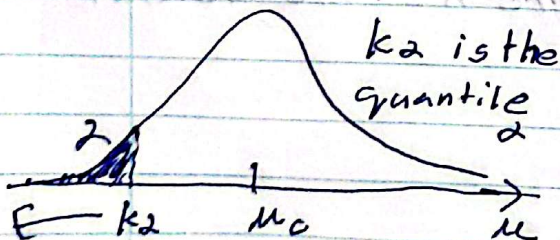
* Critical value is a cut off point that separates the rejection region from the acceptance region in hypothesis testing.

k_α is the quantile $1-\alpha$



$H_0: \mu = \mu_0 \text{ vs } H_1: \mu > \mu_0$

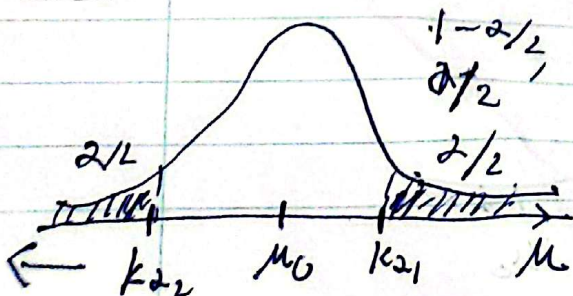
Decision

Rule: Reject H_0 if $t > k_\alpha$ 

k_α is the quantile α

$H_0: \mu = \mu_0 \text{ vs } H_1: \mu < \mu_0$

Decision

Rule: Reject H_0 if $t < k_\alpha$ 

$H_0: \mu = \mu_0 \text{ vs } H_1: \mu \neq \mu_0$

Decision

Rule = Reject H_0 if $t > k_{\alpha/2}$ or $t < -k_{\alpha/2}$

Z statistic

- * Measures how many standard deviation your sample statistic (mean or proportion) is away from the population value you are testing under the null hypothesis (H_0).

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

- * This tells you how far the sample mean is from the hypothesized mean, in terms of standard errors.
- * The larger the value of z , the stronger the evidence against H_0 .
- * Used when σ known or $n \geq 30$

4 steps in z-test

- 1) State hypotheses ($H_0: \mu = 50$, $H_1: \mu \neq 50$)
- 2) Chose significance level (0.05)
- 3) Compute z-statistic = 2.0
- 4) Find critical value (from z table) $\neq \pm 1.96$
(according to 2)
- 5) Decision $\rightarrow 2.0 > 1.96 \rightarrow$ Reject H_0 .
(There is significance evidence that $\mu \neq 0$)

13) Power of a Test

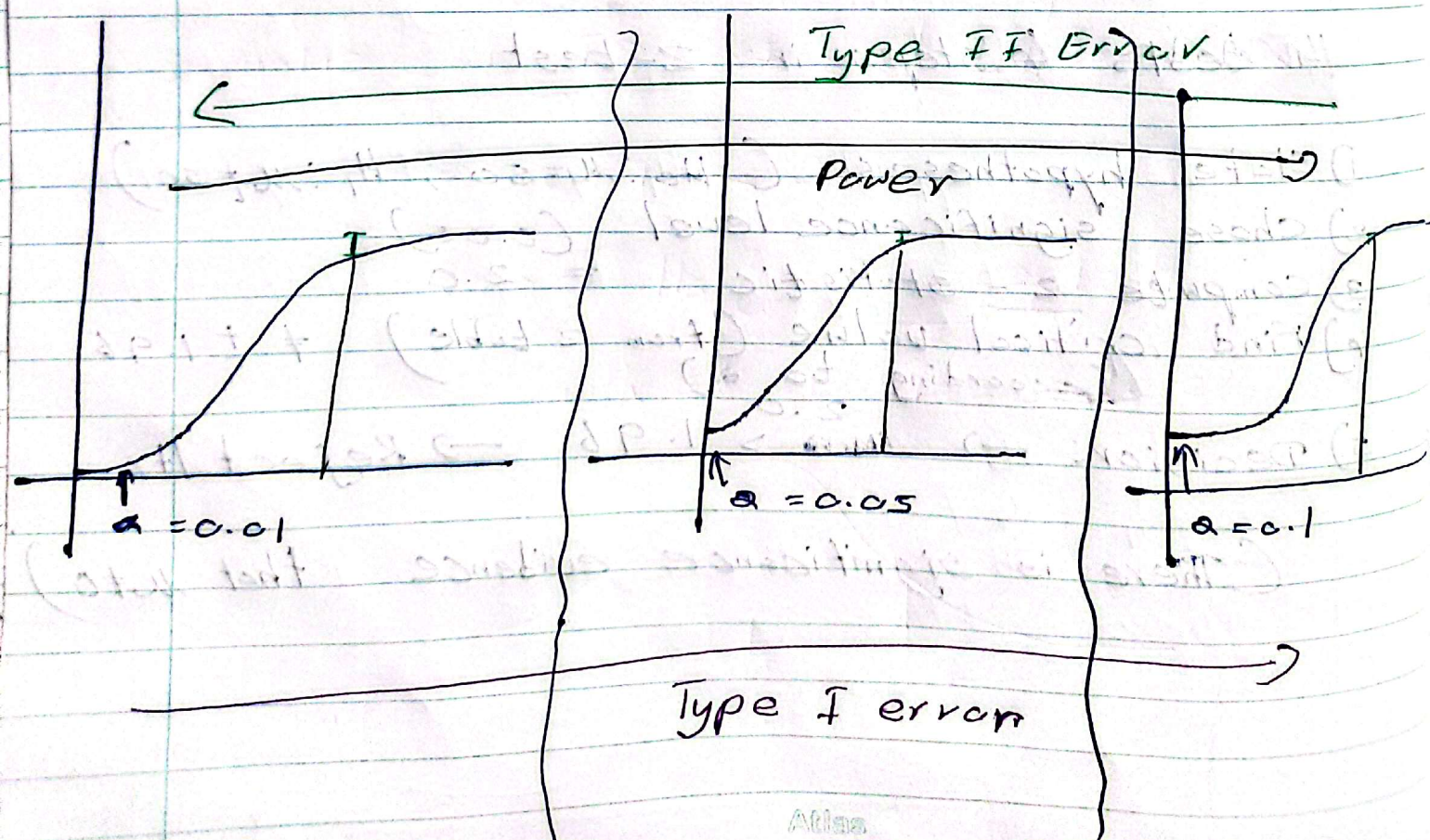
The power of a test is the probability that the test correctly rejects a false null hypothesis (H_0).

$$\text{Power} = P(\text{Reject } H_0 \mid H_0 \text{ is False})$$

High power $\Rightarrow$ more likely to detect a true effect

Low power $\Rightarrow$ higher chance of Type II error (Failing to reject H_0 , when it is false)

$$\text{Power} = 1 - \beta (\text{Type II Error})$$



14) t-distribution

* Used when σ is unknown

* Degrees of freedom (v) - Controls how heavy tails are

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

$$v = n - 1 \quad (n-1)$$

* As n increases t-distribution has a normal distribution

15) Tests for proportion

$$H_0: p = 0.5 \quad \text{vs} \quad H_1: p \neq 0.5$$

$$\hat{p} = \frac{X}{20}$$

$$Z = \frac{\frac{X}{20} - p}{\sqrt{p(1-p)}} \sqrt{20} \sim N(0, 1)$$

$$P\text{-value} = P(|Z| > |z|) = P(|Z| > 1.3616) = 0.1797$$

* when $\alpha = 0.05$ ($p > 1.797$) $\Rightarrow$ Fail to reject

16) Two sample tests

$$n_x = 10$$

$$\bar{x} = 68.442$$

$$S_x = 3.113$$

$$n_y = 9$$

$$\bar{y} = 65.949$$

$$S_y = 3.1606$$

$$t = 1.7450$$

*) $H_0: \mu_{us} - \mu_{Ar} = 0$ vs $H_1: \mu_{us} - \mu_{Ar} > 0$
 $\alpha = 0.05$

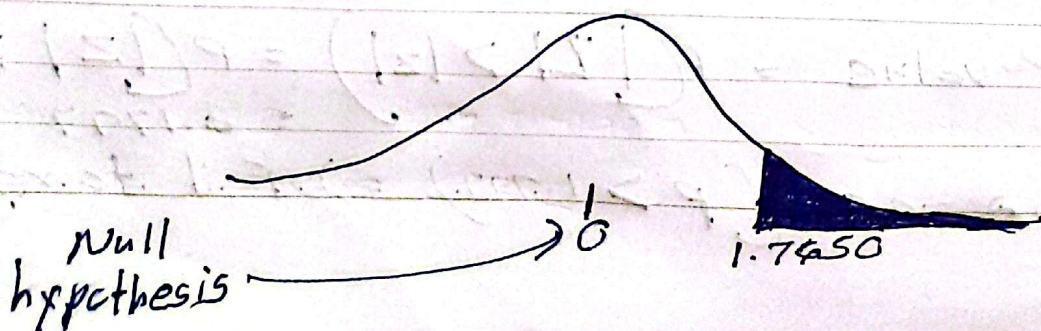
If H_0 is true :
$$T = \frac{(\bar{X} - \bar{Y}) - 0}{\sqrt{\frac{S_x^2}{10} + \frac{S_y^2}{9}}} \sim t_{14.8}$$

p-value : $P(T > 1.7450 \mid \mu_{us} = 0)$

$$= 0.0495 < 0.05$$

Reject H_0 (and accept H_1)

with 5% significance level



For proportions

$$H_0 : p_1 = p_2 \text{ vs } H_1 : p_1 \neq p_2$$

$$Z = \frac{\frac{x}{n_1} - \frac{y}{n_2} - 0}{\sqrt{\frac{(x+y)(1 - \frac{x+y}{n_1+n_2})}{n_1 n_2}}} \sim N(0,1)$$

* Right tailed :

$$P\text{-value} = P(Z > z)$$

* Left tailed :

$$p\text{-value} = P(Z < z)$$

* Two-tailed

$$p\text{-value} = P(|Z| > |z|)$$

17) Paired T-test (dependent samples)

x_i observations (sample 1) $d_i = x_i - y_i$

y_i observations (sample 2)

$$\bar{d} = \frac{\sum d_i}{n}$$

S_D = sample SD

Test
Statistic

$$T = \frac{\bar{D} - \mu_D}{S_D / \sqrt{n}} \sim t_{n-1}$$

A/B Testing

- * A/B testing is a type of experiment used to compare two versions of something - say a website, app, feature or advertisement, to see which performs better.

A (Control) $\Rightarrow$ The current / old version

B (Treatment / Variant) $\Rightarrow$ The new version you are testing.

- * A/B testing is basically a two-sample hypothesis test

H_0 : There is no difference between A and B
 $\Rightarrow H_0 : p_A = p_B$

$H_1 : p_A \neq p_B$

- 1) Collect data from both groups
- 2) Calculate a test statistic (z-score, t-score)
- 3) Find the p-value
- 4) Compare it to your significance level (0.05).

If $p < \alpha$ Reject H_0 : the difference between A and B are Statistically Significant

- ⓐ Always use ~~not~~ random assignment and enough sample size